DAY 9 | BUILD TO LEARN

# Create Flashcards, Quizzes, and Interactive HTML

Use AI as a learning designer and coding partner. Today, every important idea becomes something people can click, answer, test, and improve.

Active recall · Better questions · HTML basics · Interactive design

**Today's mission**

Turn one topic or document into a strong flashcard set, a fair quiz, and a polished standalone HTML learning activity.

Learn by building, testing, and revising

Output: one shareable HTML learning experience

WHY THIS MATTERS

# Reading feels fluent. Retrieving proves learning.

### Passive rereading

Recognition can feel like mastery even when recall is weak.

### Active recall

Trying to retrieve an answer strengthens memory and reveals gaps.

### Immediate feedback

Good explanations correct errors before they become habits.

### Iteration

Repeated practice at the right difficulty builds durable understanding.

**AI can generate practice quickly.** Your job is to make the practice accurate, fair, useful, and engaging.

THE DURABLE WORKFLOW

# One repeatable loop powers every activity today.

1
**Choose**
Select one clear learning goal and trustworthy source.

2
**Generate**
Use a precise prompt to create draft content.

3
**Check**
Verify facts, answers, level, wording, and fairness.

4
**Build**
Ask AI to package the content in HTML.

5
**Test**
Click every control and try to break the activity.

6
**Improve**
Revise from evidence, not decoration alone.

**Hands-on rule: create something every 10-15 minutes**

**Never publish an unchecked answer key**

# HTML gives a webpage its meaning and structure.

HTML stands for **HyperText Markup Language**. It uses labeled elements, called tags, to tell a browser what each piece of content is: a heading, paragraph, button, image, question, or answer.

**HTML = structure**

What is on the page?

**CSS = appearance**

How does it look?

**JavaScript = behavior**

What happens when users interact?

## Why use it?

- Opens in a web browser.
- Can work as one standalone file.
- Supports text, media, forms, buttons, and feedback.
- Easy to share locally or publish online.
- AI can help beginners create and revise it.

## HTML ANATOMY

```html
<!doctype html>
<html lang="en">
  <head>
    <title>My Quiz</title>
  </head>
  <body>
    <h1>Science Quiz</h1>
    <p>Choose an answer.</p>
    <button>Start</button>
  </body>
</html>
```

# Science Quiz

Choose an answer.

**Start**

Opening tag + content + closing tag

# Tag detective

With a partner, inspect the sample on the previous slide and identify:

1. The page title shown on the browser tab
2. The main visible heading
3. The instruction paragraph
4. The clickable control
5. The tags that contain visible content

## Then predict

```
<h2>Question 1</h2>
<p>What does HTML provide?</p>
<button>Structure</button>
```

**Challenge:** sketch what the browser will display before opening or running anything.

Time: 6 minutes

Output: labeled sketch

THREE LAYERS, THREE JOBS

H

### HTML: content and meaning

Question text, answer buttons, headings, progress, score, and feedback areas.

```
<button>Answer</button>
```

C

### CSS: visual design

Color, spacing, layout, type size, hover states, and mobile responsiveness.

```
button { color: blue; }
```

J

### JavaScript: interaction

Checks answers, flips cards, changes scores, moves forward, and resets.

```
score = score + 1;
```

**A useful AI request names all three layers:** "Create semantic HTML, clear CSS, and simple JavaScript interactions in one file."

# Ask AI for the smallest useful page

```
Create a single standalone HTML file for a learning card about [TOPIC].

Include:
- one clear heading
- a 2-sentence explanation
- three key points
- one reflection question
- clean readable CSS inside the file

Keep the code beginner-friendly. Add short comments labeling the HTML, CSS, and JavaScript sections. Do not
use external libraries.
```

**Copy prompt**

## Do it

1. Replace **[TOPIC]**.
2. Generate the code.
3. Place it in a file ending in **.html**.
4. Open it in a browser.
5. Change the heading manually.
6. Refresh and observe.

**Time: 12 minutes**

# A strong card creates one useful act of retrieval.

### One target
Test one idea, not an entire chapter.

### Clear cue
The front makes the expected answer understandable.

### Brief answer
The back is easy to check quickly.

### Useful context
Add an example, contrast, or common error where valuable.

### Mixed thinking
Use recall, explanation, application, and comparison cards.

## Weak
Front: “Explain photosynthesis.”
Back: three dense paragraphs copied from a textbook.

## Stronger
Front: “What energy source drives photosynthesis?”
Back: “Light energy, usually from the sun.”

## World-class prompt anatomy

| | |
|---|---|
| 1 **Role:** expert learning designer | 2 **Goal:** measurable learning outcome |
| 3 **Source:** evidence boundary | 4 **Mix:** varied thinking levels |
| 5 **Rules:** quality constraints | 6 **Format:** reusable output |

**Formula:** Role + Learner + Outcome + Source + Card Mix + Quality Rules + Output Format + Self-check

```
Act as an expert learning designer and subject specialist.

Create [NUMBER] high-quality flashcards for [LEARNER LEVEL] learning [TOPIC].

Learning outcome: After practice, learners should be able to [MEASURABLE OUTCOME].
Evidence boundary: Use only the source material I provide. If a fact is missing or uncertain, flag it
instead of inventing it.

Card mix:
- 30% essential recall
- 25% explain-in-your-own-words
- 20% compare/contrast
- 15% application scenarios
- 10% common misconceptions

Quality rules:
- Test one idea per card.
- Make the cue unambiguous.
- Keep answers concise but complete.
- Add a short explanation or example where useful.
```

1

### Generate 15

Use the world-class template with your topic and a trusted source.

2

### Delete 5

Remove vague, trivial, duplicate, overloaded, or unsupported cards.

3

### Improve 5

Rewrite cues, shorten answers, add examples, or raise thinking level.

## Card quality check

- ✓ One idea?
- ✓ Answerable?
- ✓ Accurate?
- ✓ Worth remembering?

## Deliverable

A final set of **10 cards** you would confidently give to a learner.

**Time: 18 minutes**

# Try retrieval before revealing.

Say your answer aloud, then click the card. Use the confidence buttons honestly; confidence helps decide what to practice next.

**Again** **Hard** **Easy**

Choose a confidence level after revealing the answer.

## What does HTML give a webpage?

Think first. Click to reveal.

PROMPT TEMPLATE | INTERACTIVE FLASHCARDS

```
Create a polished, standalone interactive flashcard web app in one HTML file.

Audience: [LEARNER LEVEL]
Topic: [TOPIC]
Cards: Use the reviewed flashcard data pasted below.

Required learning behavior:
- Show one card at a time.
- Require a click or keypress to reveal the answer.
- Include Previous, Next, Shuffle, and Reset.
- Include Again, Hard, and Easy confidence buttons.
- Track progress and confidence counts.
- Keep the learner's place if the page refreshes.

Design and accessibility:
- Mobile-friendly and readable.
- Strong color contrast and visible keyboard focus.
- Buttons have clear labels.
- Support Left/Right arrows and Space.
- Avoid external libraries and external assets.
```

## Why this works

- Names the learning behavior.
- Defines the interface.
- Requests accessibility.
- Keeps editing simple.
- Requires a testing guide.

**Copy full prompt**

# Turn your curated cards into a working HTML app.

### 1. Generate
Use the interactive flashcard prompt and paste your cards.

### 2. Open
Save as HTML and open it in a browser.

### 3. Test
Try reveal, next, previous, shuffle, reset, and keyboard controls.

### 4. Break
Refresh, click quickly, reach the last card, and test a small window.

### 5. Improve
Ask AI to fix the most important problem you observed.

**Evidence to record:** one feature that worked, one bug, one usability problem, and one improvement made.

Time: 28 minutes

Pair test before moving on

# A quiz is only useful when its questions match the learning goal.

| Question type | Best for | Watch out for |
|---|---|---|
| Multiple choice | Concept checks and diagnosis | Weak distractors and accidental clues |
| True / false | Fast misconception checks | Guessing and overly absolute wording |
| Short answer | Recall and explanation | Many valid wordings |
| Scenario | Application and decisions | Unnecessary complexity |
| Ordering / matching | Processes and relationships | Ambiguous alternatives |

### The alignment question

If the goal says "apply," but every question asks learners to "define," the quiz is misaligned.

Difficulty is not the same as confusion

### Clear stem
The question makes sense before reading the options.

### One best answer
The key is defensibly better than every distractor.

### Plausible distractors
Wrong options reflect realistic errors, not jokes.

### No hidden clues
Avoid unequal length, grammar clues, and obvious absolutes.

## Weak distractors
What does CSS control?
A. Appearance B. Bananas C. The moon D. A bicycle

## Better distractors
What is CSS mainly responsible for?
A. Page styling B. Page structure C. User interaction logic D. Server data storage

## The prompt's hidden strength

It asks AI to design the assessment **before** writing questions, and to critique the result **after** writing it.

Blueprint first

Misconception-based distractors

Answer explanations

Quality audit

Copy full prompt

```
Act as an expert assessment designer and subject specialist.

Create a [NUMBER]-question quiz for [LEARNER LEVEL] on [TOPIC].
Learning outcomes: [PASTE MEASURABLE OUTCOMES].
Evidence boundary: Use only the provided source. Flag unsupported content.

First, create a short assessment blueprint showing:
- outcome tested
- question type
- thinking level
- difficulty
- number of questions

Then write the quiz using this mix: [QUESTION MIX].

For every question:
- make the wording clear and culturally neutral
- ensure there is one defensible answer
- create plausible distractors based on likely misconceptions
- avoid trick questions, accidental clues, and unnecessary negatives
```

# Generate 10, keep 6

1. Generate a quiz using the template.
2. Answer it without looking at the key.
3. Mark unclear or disputable items.
4. Delete the weakest four.
5. Rewrite two distractors.
6. Ask a partner to challenge the answer key.

**Time: 22 minutes**

## Question audit

- ✓ Aligned to an outcome?
- ✓ One best answer?
- ✓ Plausible distractors?
- ✓ No wording clues?
- ✓ Explanation teaches?
- ✓ Verified from source?

Final output: six questions you can defend, not ten questions you merely generated.

# Which technology gives a webpage its structure?

- **HTML**
- **CSS**
- **JavaScript**
- **A spreadsheet**

Choose the best answer. Notice what useful feedback should do.

**Reset question**

PROMPT TEMPLATE | INTERACTIVE QUIZ

```
Create a polished standalone interactive quiz in one HTML file using the reviewed questions pasted below.

Learning experience:
- Start screen with title, purpose, and estimated time.
- Show one question at a time.
- Randomize question order, but keep answer logic correct.
- Give immediate explanatory feedback after each answer.
- Prevent changing an answer after feedback unless Retry is chosen.
- Show progress, score, and a final results summary.
- At the end, group missed questions by learning outcome and recommend what to review.
- Include Restart and Review Mistakes.

Design and accessibility:
- Responsive on phone and desktop.
- Keyboard accessible with visible focus.
- High contrast; do not rely on color alone.
- Clear error and success states.
- No external libraries or assets.

Code quality and safety:
```

## Upgrade ideas

- Question timer
- Hints with score cost
- Difficulty choice
- Retry missed questions
- Downloadable results
- Teacher answer-key mode

**Copy full prompt**

# Build, test, and improve your interactive quiz.

### Generate
Use six reviewed questions.

### Play
Complete the quiz normally.

### Test edges
Try zero and full scores.

### Inspect
Check feedback and scoring.

### Peer test
Watch without helping.

### Revise
Fix one content and one UX issue.

## Required evidence
Record the exact prompt used, two problems found, the revision prompt, and what changed.

## Success test
A first-time learner can understand the task, complete it, learn from mistakes, and restart without help.

**Time: 35 minutes**

**PRODUCT**

interactive quiz

**TOPIC**

HTML basics

**LEARNER**

adult beginners

**QUESTIONS / CARDS**

10

**INTERACTION**

one item at a time with feedback

**VISUAL STYLE**

clean professional classroom

**EXTRA REQUIREMENTS**

Include progress, restart, keyboard accessibility, mobile layout, and a testing checklist.

Build prompt

Copy prompt

## Your HTML creation prompt

```
Act as an expert learning designer, assessment designer, and front-end developer.

Create a standalone interactive quiz in one HTML file for adult beginners learning HTML basics.

Content: Include 10 reviewed items. Keep content accurate, clear, useful, and appropriate for the
audience. Use plausible misconception-based distractors where relevant and provide explanatory
feedback.

Learning interaction: one item at a time with feedback.
Visual direction: clean professional classroom.
Additional requirements: Include progress, restart, keyboard accessibility, mobile layout, and a
testing checklist.

Technical requirements:
- Put semantic HTML, responsive CSS, and simple JavaScript in one file.
- Use no external libraries or assets.
- Keep learning data in one clearly labeled JavaScript array.
- Add concise beginner-friendly comments.
```

# Do not regenerate everything when one part needs work.

### Bug fix
"The score becomes 7/6 after retrying. Find the cause, fix only the scoring logic, and explain the change."

### Design fix
"On phones, answer buttons are too small. Improve tap targets and spacing without changing quiz logic."

### Learning fix
"Feedback only says correct/incorrect. Add a short teaching explanation for every option."

### Accessibility fix
"Audit keyboard use, focus visibility, contrast, and button labels. Fix failures while preserving features."

**REVISION FORMULA** `Current behavior + expected behavior + evidence/example + scope to preserve + requested verification`

# Pick one real weakness

Choose the most important issue from peer testing:

Incorrect content · Confusing question · Broken interaction · Poor feedback · Mobile problem · Accessibility problem

```
In the attached HTML, [CURRENT PROBLEM].

Expected behavior: [WHAT SHOULD HAPPEN].
Evidence: [HOW I OBSERVED THE PROBLEM].

Fix this issue while preserving [WORKING FEATURES]. Return the complete updated HTML.
Then explain the cause, the exact change, and how I should test the fix.
```

## Test the revision

1. Confirm the original issue is fixed.
2. Retest nearby features.
3. Compare before and after.
4. Keep the change only if it improves the experience.
5. Document what you learned.

**Time: 18 minutes**

### Content

- Facts and keys verified
- Level and language fit
- No duplicates or ambiguity
- Feedback teaches

### Behavior

- Every button works
- Scores remain correct
- Restart resets everything
- First and last items work

### Experience

- Instructions are clear
- Keyboard works
- Focus is visible
- Phone layout is usable

## Edge cases worth trying

0% score · 100% score · Very long answer · Rapid clicks · Refresh halfway · Narrow screen

## Golden rule

Test what users **might** do, not only what you hope they do.

### Verify answers

AI can confidently produce incorrect keys and explanations.

### Protect data

Do not paste private student records or confidential materials into unapproved tools.

### Design fairly

Avoid stereotypes, inaccessible design, cultural assumptions, and trick wording.

### Be transparent

State when AI helped create materials and who reviewed them.

**Human responsibility does not shrink because generation becomes faster.**

### Card repair
Improve ten weak AI flashcards.

### Misconception quiz
Build questions around common errors.

### Scenario sprint
Turn definitions into application decisions.

### Accessibility audit
Fix keyboard, focus, contrast, and labels.

### Theme remix
Change visual style without breaking logic.

### Peer challenge
Swap apps and find three improvements.

### Question ladder
Create easy, medium, and hard versions.

### Explain errors
Write feedback for every distractor.

### Source lock
Generate only from a trusted passage.

### Mobile test
Improve a cramped phone layout.

### Code detective
Find where title, colors, and data live.

### Remix
Convert quiz content into flashcards.

# Move quickly from understanding to repeated creation.

| Time | Block | Visible learner output |
|---|---|---|
| 00:00-00:30 | Active learning and HTML basics | First learning-card HTML page |
| 00:30-01:15 | Flashcard design and generation | Ten curated flashcards |
| 01:15-01:25 | Break | Reset |
| 01:25-02:05 | Interactive flashcard build | Tested flashcard app |
| 02:05-02:50 | Quiz design and question audit | Six defensible quiz questions |
| 02:50-03:00 | Break | Reset |
| 03:00-03:40 | Interactive quiz build and peer test | Working quiz plus test notes |
| 03:40-04:00 | Improvement sprint and showcase | Revised capstone and reflection |

# Build one HTML activity another person can learn from without your help.

**Goal**
One measurable learning outcome.

**Content**
Verified cards or questions.

**Interaction**
Useful feedback and progress.

**Quality**
Peer tested and revised.

**Reflection**
Explain choices and limits.

**Celebrate completion** **Print / Save PDF**

## Showcase script

1. Who is this for?
2. What should they learn?
3. What did AI generate?
4. What did you change?
5. What problem did testing reveal?
6. What would you improve next?

# AI makes the first draft fast. Your judgment makes it worth using.

Define the goal | Prompt precisely | Verify content | Test behavior | Improve from evidence

## Exit ticket

1. What can HTML help you create?
2. What makes a strong flashcard?
3. What makes a fair quiz question?
4. What did testing change in your project?
5. What will you build next?